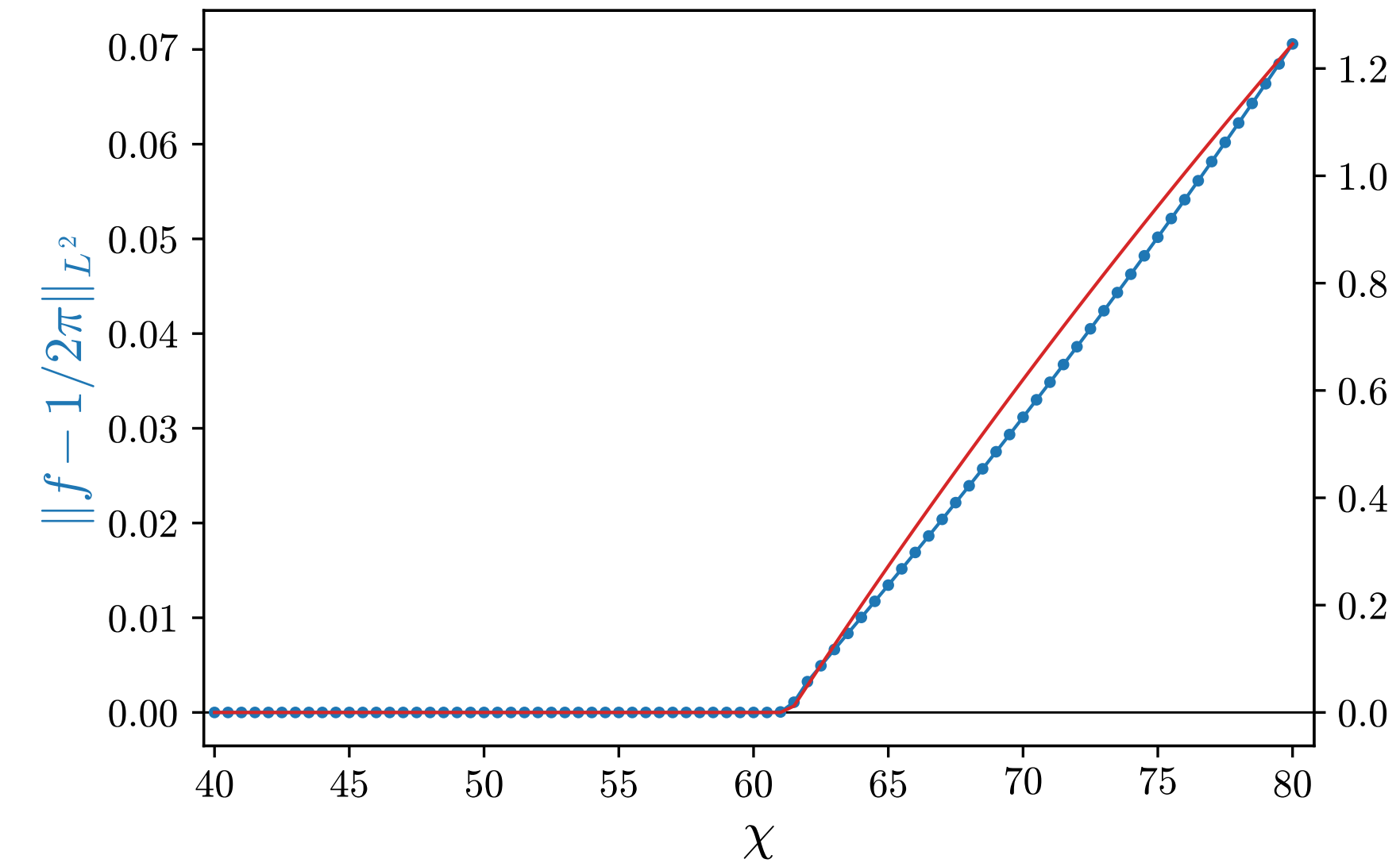


$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

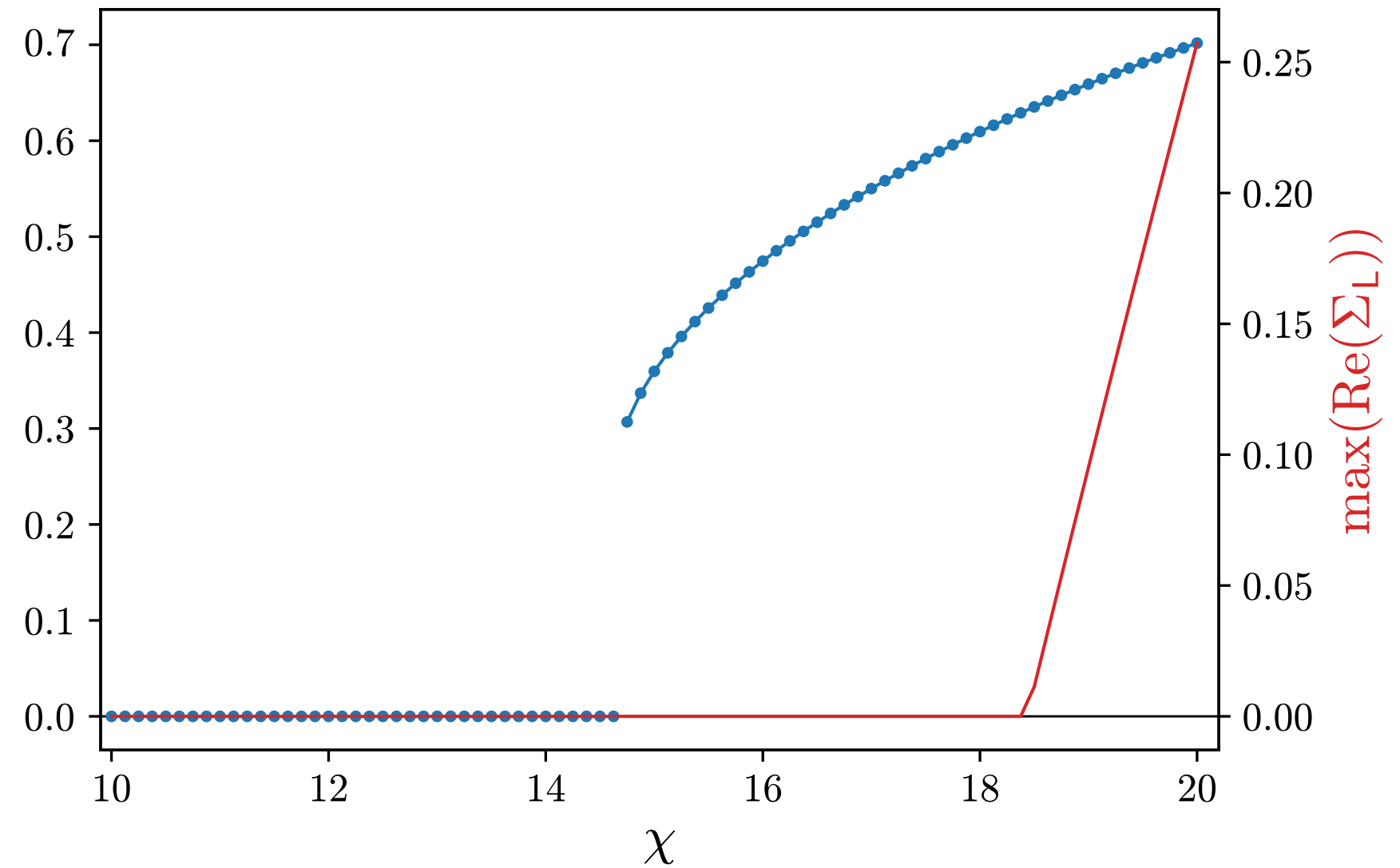
- What happens as $t \rightarrow \infty$?

• What is a super-critical transition?

Existence at or before loss of linear stability



$\tau < \tau^*$



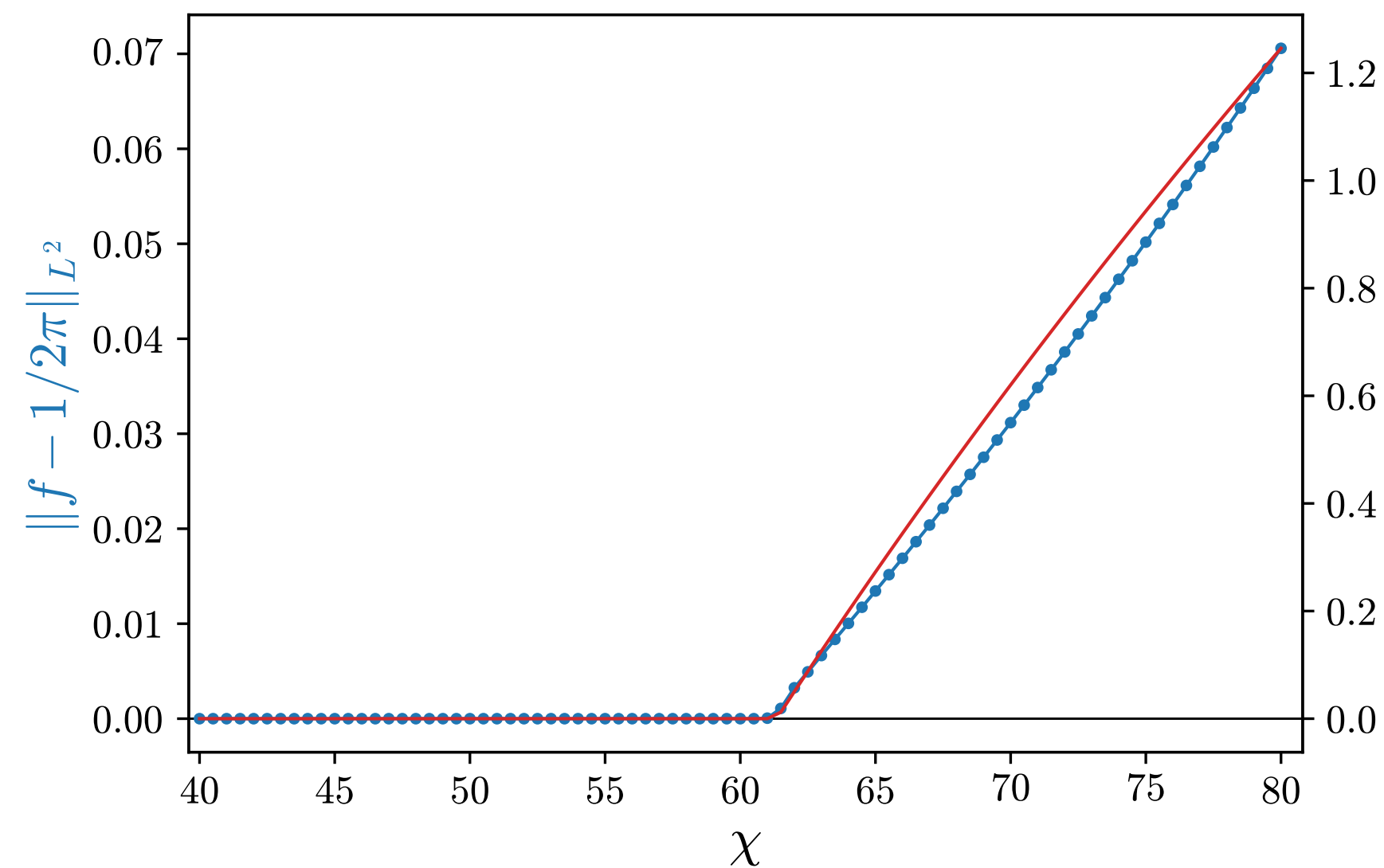
$\tau > \tau^*$

Antdynamics

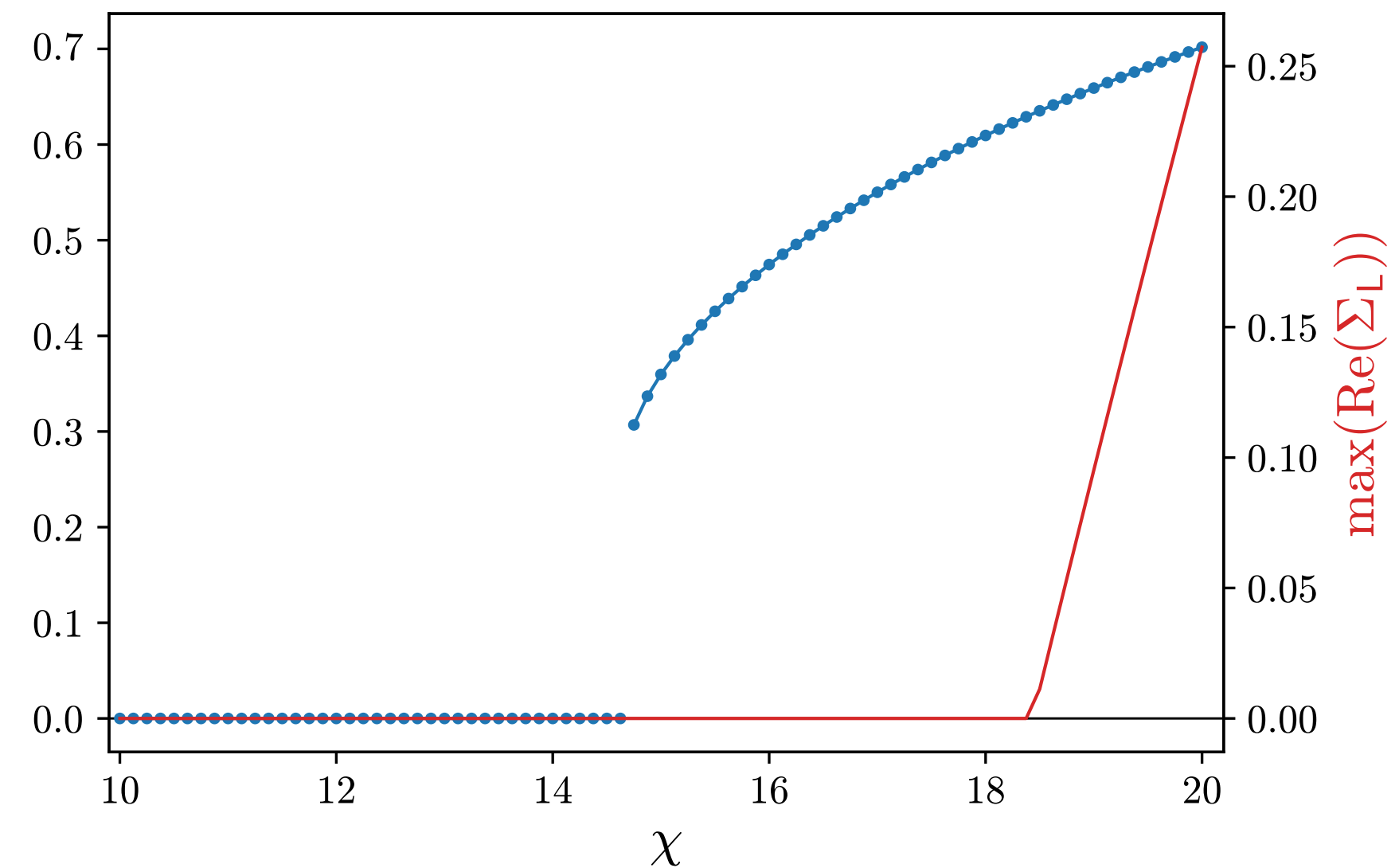
Ant dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?
- What is a super- to subcritical transition?
 - Existence *at or before* loss of local linear stability



$\tau < \tau^*$



$\tau > \tau^*$

Ant metastability? (Open)